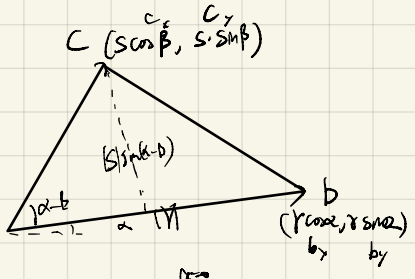
$(b-a) \times (c-a)$



$$\begin{vmatrix} b_x & b_y \\ c_x & c_y \end{vmatrix} = b_x c_y - c_x b_y = r s \cos\alpha \sin\beta - s\cos\beta\, r\sin\alpha$$

$$= rs(\sin\beta\cos\alpha - \cos\beta\sin\alpha)$$

$$= rs\sin(\alpha-\beta) \quad = 2\,\text{area}_{\Delta ABC} \text{ with oriented}$$

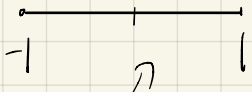
$\alpha-\beta>0$ ↺

$\alpha-\beta<0$ ↻

---

fun gauss_quad copy... But actually:

$$\int_{-1}^{1} f(s)\,ds = \sum_{i=1}^{n} f(s_i) w_i$$



$$J = \begin{bmatrix} 0 & b_1 & & \\ b_1 & \ddots & \ddots & 0 \\ & \ddots & \ddots & b_{n-1} \\ 0 & & b_{n-1} & 0 \end{bmatrix} \qquad b_i = \frac{i}{\sqrt{4i^2-1}}$$

$$J v_i = \lambda_i v_i \qquad v_i^T v = 1$$

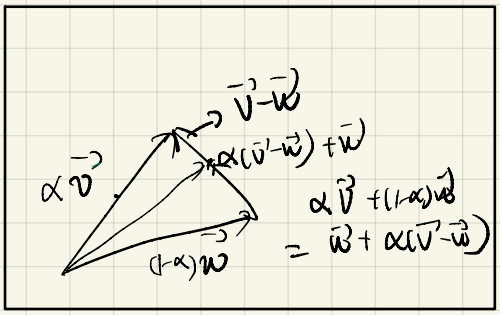
$(\lambda_1 \cdots \lambda_n)$ ← points $\qquad w_i = 2(v_i[1])^2$

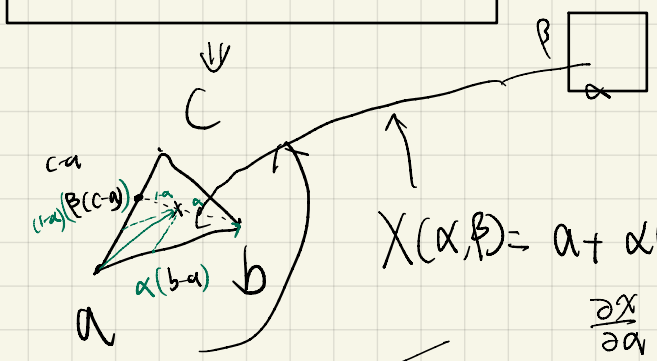
$$x = 2q-1 \qquad q = \frac{x+1}{2}$$

$$\int_0^1 f(q)\,dq = \int_{-1}^{1} f(q(x))\,dq(x) = \int_{-1}^{1} f\left(\frac{x+1}{2}\right)\frac{1}{2}dx = \frac{1}{2}\int_{-1}^{1} f\left(\frac{x+1}{2}\right)dx$$

$$= \sum \frac{1}{2} f\left(\frac{\lambda_i+1}{2}\right) w_i = \sum f(q_i)\cdot w_i'$$

$$q_i = \frac{\lambda_i+1}{2} \qquad w_i' = \frac{1}{2} w_i$$



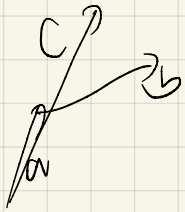


$\alpha, \beta \in [0,1] \times [0,1]$

$$X(\alpha,\beta) = a + \alpha(b-a) + (1-\alpha)\beta(c-a) = \begin{bmatrix} x \\ y \end{bmatrix}(\alpha,\beta)$$

$$\frac{\partial x}{\partial \alpha} = (b_x - a_x) - \beta(c_x - a_x) \qquad \frac{\partial x}{\partial \beta} = (1-\alpha)(c_x - a_x)$$

$$\frac{\partial y}{\partial \alpha} = (b_y - a_y) - \beta(c_y - a_y) \qquad \frac{\partial x}{\partial \beta} = (1-\alpha)(c_y - a_y)$$



$$\det \begin{bmatrix} b_x - a_x - \beta(c_x - a_x) & (1-\alpha)(c_x - a_x) \\ b_y - a_y - \beta(c_y - a_y) & (1-\alpha)(c_y - a_y) \end{bmatrix}$$

$$J = (1-\alpha)\left|\det\left[(\vec{b}-\vec{a}) - \beta(\vec{c}-\vec{a}),\ (\vec{c}-\vec{a})\right]\right| = D$$

$(1-\alpha) \geq 0$; by det column=0

$$\int f(x,y)\,dx\,dy = \int f(X(\alpha,\beta))\frac{\partial(x,y)}{\partial\alpha\,\partial\beta}\,d\alpha\,d\beta = \int f(X(\alpha,\beta))\,J\,d\alpha\,d\beta$$

$$\overset{\text{f is regular}}{\overset{\text{Fubini}}{=}} \int_\beta \int_\alpha f(X(\alpha,\beta))(1-\alpha)D\,d\alpha\,d\beta =$$

$$\int f(x)dx = f(x(h))\,x'(h)\,dh$$

$$\text{Approx} = D\int_\beta \sum_{i=1}^{n} f(X(\alpha_i,\beta))(1-\alpha_i)w_i\,d\beta$$

$$= D\sum_{j=1}^{n}\sum_{i=1}^{n} f(X(\alpha_i,\beta_j))(1-\alpha_i)w_i w_j$$

$$= \sum w_k f(\bar{x}_k)$$

$\bar{w}_k = D(1-\alpha_i)w_i w_j \quad \bar{x}_k = X(\alpha_i,\beta_j)$

$\alpha_i, w_j \quad \beta_i, w_j$ from λ_i, w_i

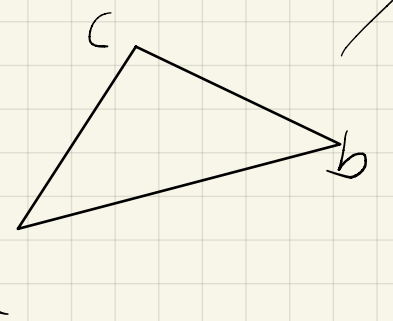
If we want p-th order accuracy, since we multiply $(1-\alpha)$

we need $(p+1)$ order accuracy for each α_i and β_i

For each quadrilateral_quadrature ( )

① Check mesh

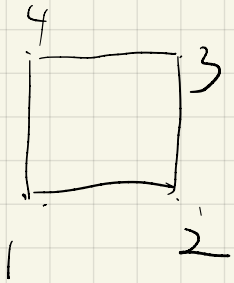
② Return (x_i, y_i) w_i



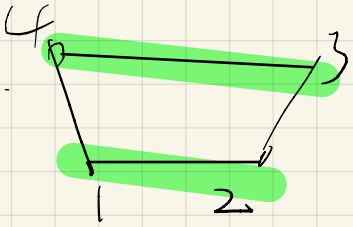
for $\alpha_i = (a_1 \cdots a_n)$, $\beta_i = (a_1 \cdots a_n)$

$(x_i, y_i)[\alpha_i][\beta_i] = a + \alpha_i (b-a) + (1-\alpha_i)\beta_i (c-a)$

Check mesh Just need decide 2 line

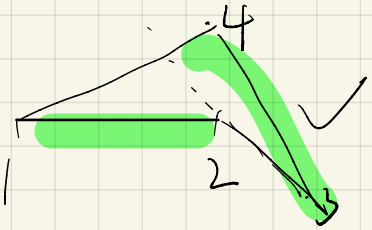


(1)

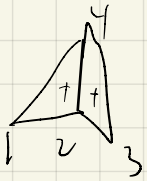


+ +
+ +

(2)

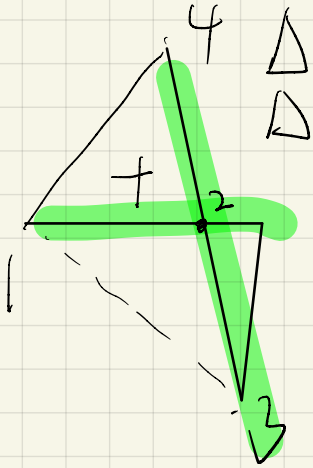


△ 123 −
△ 134 +



△124 +
△234 +

(3)

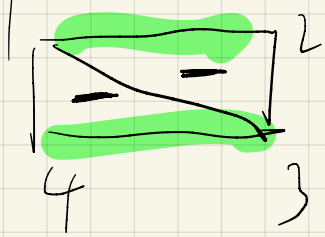


△123 −
△134 +

△ 124 −
△ 234 0

throw error!

(4)



throw error

if 2 triangle which are +

Then we pass check and select 2 △ to compute.